Assignment 3

Total = 10 Marks

Multiple Choice Questions

Question 1:

A discrete random variable taking values obtained by counting the number of occurrences of an event in a sequence of independent trials until the first success occurs is modelled by which distribution?

(a) Binomial Distribution
(b) Geometric Distribution
(c) Hypergeometric Distribution
(d) Poisson Distribution

[1 *Mark*]

Correct answer: (b) Geometric Distribution

Detailed Solution:

The probability that the first success of a Bernoulli trial occurs on the xth trial can be found using the geometric distribution. Here, the random variable is the number of trials (X) in the sequence of a Bernoulli process to get the first occurrence.

Question 2:

Which continuous probability distribution is positively skewed, takes only non-negative values, and can be treated as the sum of exponentially distributed random variables?

(a) Uniform Distribution
(b) Gamma Distribution
(c) Normal Distribution
(d) Beta Distribution

[1 *Mark*]

Correct answer: (b) Gamma Distribution

Detailed Solution:

Gamma Distribution is a continuous probability distribution that is positively skewed over the positive side of the real line. The random variable (X) is continuous and positively skewed that takes non-negative values only. It can be treated as the sum of exponentially distributed random variables each with the same parameter.

Question 3:

A random variable X follows a log-normal distribution if:

(a) X^2 follows a normal distribution
(b) e^X follows a normal distribution
(c) $\ln(X)$ follows a normal distribution
(d) $1/X$ follows a normal distribution

[1 *Mark*]

Correct answer: (c) $\ln(X)$ follows a normal distribution

Detailed Solution:

Log–normal distribution is a continuous probability distribution used for variables that take only positive values. A random variable X is said to follow a log-normal distribution if its logarithmic transformation (Y) follows a normal distribution, i.e., $Y = ln(X)$.

Question 4: A water security analyst is modeling the low-flow characteristics of a river to ensure sustainable domestic water supply during the dry season. Which extreme value probability distribution is most appropriate for assessing the risk of severe drought?

(a) Extreme Value Type-I (Gumbel) Distribution
(b) Extreme Value Type-III (Weibull) Distribution
(c) Normal Distribution
(d) Log-Normal Distribution

[1 *Mark*]

Correct answer: (b) Extreme Value Type-III (Weibull) Distribution

Detailed Solution:

Extreme Value Type-III (Weibull) Distribution, also known as Weibull distribution, is utilized for the extremes in the direction towards which the parent distribution is limited. It is generally used for modelling minimum values in engineering. The random variable (X) is continuous and is the minimum of a sample from an asymmetric distribution and takes non-negative values. Since river discharge cannot be negative, low-flow values are naturally bound below by zero. Therefore, the Weibull distribution is appropriate for modeling minimum river flows and assessing the risk of severe drought.

Question 5: Let X be a discrete random variable with the following probabilities: $P(X = 0) = 0.15,\ P(X = 1) = 0.25,\ P(X = 2) = 0.40,\ P(X = 3) = 0.15,\ P(X = 4) = 0.05$. What is $P(X \leq 2)$?

(a) 0.40
(b) 0.65
(c) 0.80
(d) 0.85

[1 *Mark*]

Correct answer: (c) 0.80

Detailed Solution:

$$P(X \leq 2) = P(X = 0) + P(X = 1) + P(X = 2) = 0.15 + 0.25 + 0.40 = 0.80$$

Question 6:

A machine learning model correctly identifies 70% of anomalies ($p = 0.70$). If 5 anomalies occur, what is the probability that the model correctly identifies exactly 4 of them? (Use Binomial distribution)

(a) 0.36
(b) 0.24
(c) 0.17
(d) 0.30

[2 *Mark*]

Correct Answer: (a) 0.36

Detailed Solution:

In this case, $p = 0.70, n = 5\ and\ x = 4$

For Binomial Distribution,

$$p_X\ (x; n,\ p) = {}^{n}C_x\ p^x\ (1\ -\ p)^{n-x} = {}^{5}C_4\ 0.70^4\ (1\ -\ 0.70)^{5-4}$$
$$= \frac{5!}{4! \times (5-4)!} \times 0.70^4\ \times (1\ -\ 0.70)^1 = 5 \times 0.2401 \times 0.30 = 0.36$$

Therefore, the probability that the machine learning model correctly identifies exactly 4 of the anomalies is 0.36.

Question 7:

Considering the time required (X) (in minutes) to train a single epoch of a deep neural network on a cloud GPU cluster follows a normal distribution with a mean of 10 minutes and a standard deviation of 5 minutes, determine the probability that a given epoch takes between 20 minutes and 25 minutes to train?

(a) 0.02
(b) 0.05
(c) 0.10
(d) 0.20

[3 *Mark*]

Correct answer: (a) 0.02

Detailed Solution:

Using standard normal distribution, we can transform the limits of X to limits of Z and then use standard normal tables.

Given: μ= 10 min and σ = 5 min

$x = 20$ transforms to $z = \frac{x-\mu}{\sigma} = \frac{20-10}{5} = \frac{10}{5} = 2$

Similarly, $x = 25$ transforms to $z = \frac{x-\mu}{\sigma} = \frac{25-10}{5} = \frac{15}{5} = 3$

From standard normal distribution table, the values of $F_Z(2)$ and $F_Z(3)$ are :

$$F_Z(2) = 0.977 \text{ and } F_Z(3) = 0.998.$$

Thus, the probability is,

$$P(20 \leq X \leq 25) = P(2 \leq Z \leq 3) = F_Z(3) - F_Z(2) = 0.998 - 0.977 = 0.021$$

So, the probability that a given epoch takes between 20 minutes and 25 minutes to train is 0.02.